

Pascal Sikorski

+1 (314)-255-4735 | pascalos@engineering.upenn.edu | pascalsikorski.com | linkedin.com/in/pascalsikorski

EDUCATION

University of Pennsylvania

Ph.D. in Computer and Information Science

Advisors: Vijay Kumar and George Pappas

Philadelphia, PA

August 2026 – May 2031 (Expected)

Saint Louis University

B.S. Double Major in Computer Science and Mathematics; Minor in Philosophy

GPA: 3.94/4.00

Saint Louis, MO

August 2022 – May 2026

EXPERIENCE

Saint Louis University - Mecharithm Lab

Lead Robotics Researcher

Saint Louis, MO

January 2024 – May 2026

- Led a 6-person team to build an open-source speech-to-motion control tool, utilizing a speech/LLM edge-computing system to communicate spoken commands over TCP to a Raspberry Pi Pico W and Arduino unit, achieving 85% command-execution accuracy for mobile and manipulator systems, resulting in 2 IEEE publications.
- Built a self-driving multi-robot wheelchair system in ROS2 utilizing a custom ORB-SLAM3 port, Nav2, and MoveIt for assistive driving/pick-and-place tasks. Incorporated an EKF to fuse SLAM odometry, IMU, and GPS for global localization, resulting in 0 collisions across multiple tests in a cluttered 450 square foot room.
- Developing a multimodal transformer model and ROS2 bridge to decode user intent from EEG, EMG, and eye-tracking data to create assistive robotic actions. Utilizing YOLO, NVIDIA FoundationPose, and MoveIt, resulting in a current 92% intent read to pick-and-place success rate.
- Collaborated on research with Prof. Madi Babaiaasl, and helped mentor 7+ students in research.

Saint Louis University - Computer Science Department

Teaching Assistant

Saint Louis, MO

August 2023 – May 2026

- Assisted in teaching MENG 4930/5930 - Modern Robotics with Prof. Madi Babaiaasl. (1 term)
- Assisted in teaching CSCI 3100 - Algorithms with Prof. Michael Goldwasser. (1 term)
- Assisted in teaching CSCI 2100 - Data Structures with Prof. Michael Goldwasser. (2 terms)
- Assisted in teaching CSCI 1060 - Scientific Programming with Prof. David Wisbey. (2 terms)
- Guided and educated a total of 210+ students.

Oregon State University - Intelligent Machines and Materials Lab

Robotics Research Intern

Corvallis, OR

June 2025 – August 2025

- Created a UR5e arm trajectory caching package in ROS2 for apple harvesting tasks by precomputing optimal paths using a multi-objective cost optimizer (considering length and manipulability) via RRT* and MoveIt. Trajectories were locally saved for offline deployment, reducing on-site compute.
- Built an optical-flow odometry estimator using Shi-Tomasi feature detection and Lucas-Kanade in OpenCV to estimate vertical motion while reducing localization drift within orchards.
- Extended OSU's SeeTree orchard localization tool by piping detected trunks into parameterized MoveIt collision objects. Fused optical-flow odometry with yaw to estimate 6D tree poses, providing obstacle-aware planning.
- Collaborated on research with Prof. Joseph Davidson through the Robots in the Real World REU.

California Institute of Technology - AMBER Lab

Robotics Research Intern

Pasadena, CA

June 2024 – August 2024

- Successfully implemented untested theory of a Belief CBF control filter in ROS2, utilizing SLAM for safe planning/control given an uncertain/probabilistic odometry or map.
- Evaluated the Belief CBF filter on TurtleBot3 mobile robots utilizing LiDAR-based SLAM in Gazebo simulation.
- Collaborated on research with Prof. Aaron Ames and Dr. Xiao Tan through The Leadership Alliance SR-EIP.

HONORS AND AWARDS

National Science Foundation Graduate Research Fellowship

Awarded the NSF Graduate Research Fellowship to support doctoral studies (\$159,000 total).

2026-2029

Edward A. Doisy Award

Sole recipient among graduating SLU science students, recognizing leadership and academic excellence.

2026

Saint Louis University Statistics Award

Department prize for top statistics performance; one of 2 SLU students to be faculty-nominated this cycle.

2026

Goldwater Scholar

Most prestigious undergraduate STEM award in the US. Selected for demonstrating exceptional research.

2025

Pi Mu Epsilon Inductee

National math honor society; one of 2 SLU students to be faculty-nominated this cycle.

2025

Saint Louis University SSE Research Travel Grant

Competitive SLU School of Science and Engineering travel grant to present at IEEE BigData.

2024

Saint Louis University Calculus Award

Department prize for top calculus performance; one of 3 SLU students to be faculty-nominated this cycle.

2024

Drahmann Research Sciences Award

Earned independent research funding awarded by a SLU research-faculty committee.

2023

PUBLICATIONS AND RELATED WORK

Journal Articles

- [1] Tipu Sultan, Guangping Liu, **Pascal Sikorski**, Samah Alshathri, Walid El-Shafai, and Madi Babaiasl. “NeuroFusion-Trans: A Novel Transformer-Based EEG-EMG Fusion Model for Assistive Robotics.” *IEEE Access*, June 2025, vol. 13, pp. 103245-103263.
- [2] Tipu Sultan, Guangping Liu, **Pascal Sikorski**, and Madi Babaiasl. “TransNN-MHA: A Transformer-Based Model to Distinguish Real and Imaginary Motor Intent for Assistive Robotics.” *IEEE Access*, May 2025, vol. 13, pp. 83650-83673.

Conference Publications

- [3] **Pascal Sikorski**, Leendert Schrader, Kaleb Yu, Lucy Billadeau, Jinka Meenakshi, Naveena Mutharasan, Flavio Esposito, Hadi Ali Akbarpour, and Madi Babaiasl. “Deployment of Large Language Models to Control Mobile Robots at the Edge.” *Proceedings of the Third International Conference on Mechatronics, Control, and Robotics (ICMCR 2025)*. Singapore, Feb. 14-16, 2025, pp. 19-24.
- [4] **Pascal Sikorski**, Kaleb Yu, Lucy Billadeau, Flavio Esposito, Hadi Ali Akbarpour, and Madi Babaiasl. “Improving Robotic Arms through Natural Language Processing, Computer Vision, and Edge Computing.” *Proceedings of the Third International Conference on Mechatronics, Control, and Robotics (ICMCR 2025)*. Singapore, Feb. 14-16, 2025, pp. 35-41.
- [5] Baodong Chen, Wei Wang, **Pascal Sikorski**, and Ting Zhu. “Adversary is on the Road: Attacks on Visual SLAM using Unnoticeable Adversarial Patch.” *Proceedings of the Thirty-Third USENIX Security Symposium (USENIX 2024)*. Philadelphia, Pennsylvania, Aug. 14-16, 2024, pp. 6345-6362.

Poster Presentations

- [6] **Pascal Sikorski**, Tate Severs, Jacob Verschueren, and Jason Knouft. “Socioeconomic Patterns of Water Contamination in St. Louis Water Bodies.” *Presented at the SSE Undergraduate Student Showcase*. Saint Louis, MO, Apr. 30, 2025.
- [7] Joel Aikkarakudiyil Joby, **Pascal Sikorski**, Tipu Sultan, Hadi Akbarpour, Flavio Esposito, and Madi Babaiasl. “EMG-TransNN-MHA: A Transformer-Based Model for Enhanced Motor Intent Recognition in Assistive Robotics.” *Presented at IEEE International Conference on Big Data (IEEE BigData 2024)*. Washington D.C., Dec. 15-18, 2024, pp. 8691-8693.

Presentations at Professional Meetings

- [8] **Pascal Sikorski**, Xiao Tan, and Aaron Ames. “Control Barrier Functions Through SLAM: A Probabilistic Approach.” *Presented at The Leadership Alliance National Symposium*. Hartford, Connecticut, Jul. 27, 2024.

ACTIVITIES

MORPH

Saint Louis, MO

K-12 Educational Robotic Kit Founder

August 2025 – July 2026

- Prototyped a \$400 robotic kit, *MORPH*, to introduce modern robotics, including control, perception, SLAM, and general multi-purpose AI to under-resourced K-12 Saint Louis school districts, securing \$3000+ in funding.
- Led a team of 4 to develop hardware and ROS2 software, with our first prototype supporting differential-drive and LiDAR, creating a plug-and-play learning option intended for novice audiences.
- Working with a local non-profit, the Unity Foundation, to deploy our curriculum to school districts.

Saint Louis University Competitive Programming Team

Saint Louis, MO

Competitive Programmer

August 2022 – May 2026

- Engaged in ICPC tournaments to apply and test programming skills across various algorithm-based problems.
- Participated weekly with Prof. Michael Goldwasser to expand knowledge on algorithms and data structures.
- Recent competitions in Feb. 2023, Feb. 2024, Nov. 2024, and Nov. 2025.

Blizzard Entertainment

Saint Charles, MO

Overwatch Professional Esports Player

April 2018 – July 2023

- Practiced daily with 5 teammates and self-studied to achieve top global performance.
- Competed with the Polish national team for the 2023 Overwatch World Cup.
- Achieved a competitive ranking of 6th best in the Americas.
- Continuing to apply strong team-based communication and problem-solving skills to applied research.

SKILLS

Robotics: ROS2, SLAM, Autonomous Navigation, Safe Control, Pose Estimation, MoveIt, URDF, Isaac Sim, Gazebo.

Software: Docker, DDS, CMake, colcon, Git, Bash, OpenCV, Ubuntu/Linux.

Programming: Python, C/C++, Java, R, MATLAB.

Hardware: LiDAR, RGB-D, IMU, GPS, Edge Devices, Mobile Robots, Manipulators, Raspberry Pi, Arduino, Circuitry.

ML: Neural Networks, Reinforcement Learning, Transformer Models, Multimodal Learning, PyTorch, YOLO.

Languages: Native English, Intermediate Polish, Beginner French.

RESEARCH INTERESTS

Interests: Field Robotics, Collaborative/Multi-Robot Planning, Human-Robot Interaction, Game Theory.